



DISASTER SHIELD AI

Real-Time Disaster Intelligence · AI-Powered Early Warning · Community Survival Platform

"When every second counts, intelligence saves lives."



GLOBAL

Disaster Coverage



REAL-TIME

Live Alert Feeds



AI-DRIVEN

Risk Intelligence

THE PROBLEM

Disasters Are Getting Worse. And Responses Are Still Too Slow.

90,000+

Deaths per year from
natural disasters globally

\$300B+

Economic losses annually
from climate disasters

72 hrs

Critical window where
early warning saves lives

Current Gaps

- Fragmented information: no single source for real-time disaster data
- Delayed alerts: communities receive warnings too late to evacuate safely
- No integrated survival guidance: people don't know what to do when disaster strikes
- Limited coordination between emergency services, citizens, and aid agencies

Disaster Shield AI — One Platform. Total Situational Awareness.



Live GIS Map



Real-time global disaster events on an interactive map with severity layers



AI Risk Analysis

Machine learning models predict disaster spread, risk zones, and impact



Instant Alerts

Push notifications for early disasters with actionable guidance



Survival Academy

Curated video library from Red Cross, FEMA, CAL FIRE & NatGeo



Weather Intelligence



Hyperlocal weather feeds integrated with disaster threat assessment



Facilities Finder

Nearest hospitals, shelters and emergency services at your fingertips

What Makes Disaster Shield AI Different

□ Real-Time GIS Disaster Map

- Google Maps integration with live disaster event overlays
- Severity-coded markers: Critical, High, Medium, Low
- Glassmorphic InfoWindow popups with full disaster telemetry
- Impact radius visualization and coordinate tracking

□ Survival Academy

- 10+ verified video lessons from Red Cross, FEMA & CAL FIRE
- Filter by disaster type, difficulty, and category
- Interactive modal with step-by-step emergency guides
- Emergency checklists, dos & don'ts for every scenario

□ AI-Powered Intelligence

- AI assistant for real-time disaster Q&A and survival advice
- Predictive risk zone modeling based on historical patterns
- Automated severity classification with ML confidence scores
- Natural language disaster briefings and threat summaries

□ Alerts & Community Response

- Location-aware push notifications for nearby disasters
- Community reporting with photo and severity data
- Nearest shelters, hospitals, and emergency contacts
- Offline-ready emergency checklists and evacuation routes

Built on a Modern, Scalable Tech Stack

FRONTEND	React 18 + Vite Framer Motion · Lucide Icons · Google Maps API · Recharts
BACKEND	Node.js + Express RESTful API · JWT Auth · Mongoose ODM · Nodemon
DATABASE	MongoDB Atlas Cloud NoSQL · Geospatial queries · Aggregation pipelines
AI & DATA	Google Gemini AI Natural language disaster Q&A · Risk analysis · Threat classification
SERVICES	3rd-Party APIs OpenWeatherMap · ReliefWeb · GDACS · YouTube Data API

Who Benefits from Disaster Shield AI?

□ □ □ □ General Public

- Get real-time alerts for nearby disasters
- Access survival guides & evacuation routes
- Find nearest shelters, hospitals & safe zones

□ □ Government & NDRF

- Monitor disaster events across regions in real-time
- Coordinate emergency response operations
- Deploy resources based on AI-predicted impact zones

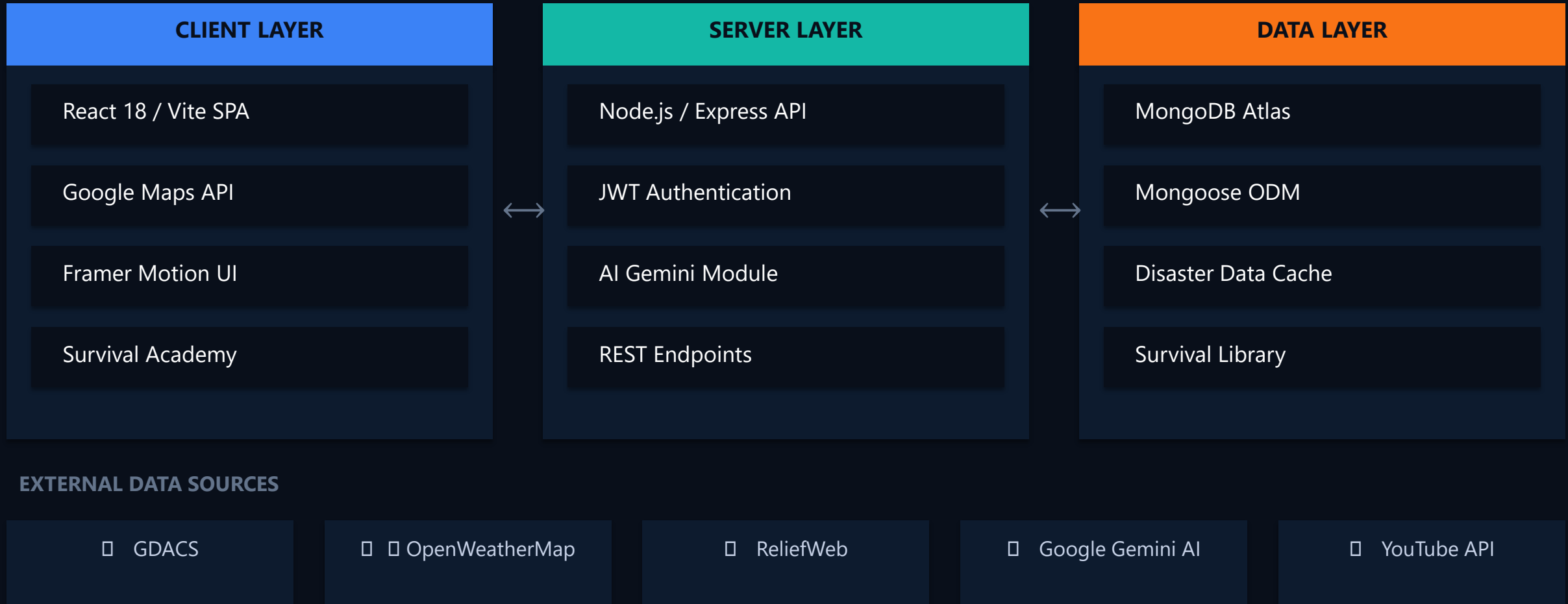
□ Media & Journalists

- Real-time verified disaster data for reporting
- GIS-mapped incident locations and severity
- Historical trends and disaster analytics

□ Schools & NGOs

- Disaster preparedness training via Survival Academy
- Community awareness and drill planning
- Educational resources verified by Red Cross & FEMA

System Architecture Overview



What's Next for Disaster Shield AI

Phase 1 COMPLETE

- ❑ Real-time GIS disaster map
- ❑ Google Maps integration
- ❑ Survival Academy video library
- ❑ AI-powered disaster assistant
- ❑ Weather intelligence module

Phase 2 IN PROGRESS

- ❑ Mobile app (React Native)
- ❑ Community disaster reporting
- ❑ Push notification system
- ❑ Offline emergency mode
- ❑ Multi-language support

Phase 3 PLANNED

- ❑ Satellite imagery integration
- ❑ Drone coordination module
- ❑ Government API partnerships
- ❑ Predictive AI model (v2)
- ❑ Global NGO data network



DISASTER SHIELD AI

Protecting Communities. Saving Lives. Powered by Intelligence.

□ Global Coverage

□ Real-Time Alerts

□ AI-Powered

□ Survival Training

Thank You

Questions & Demo

github.com/Disaster-Shield-AI

TEAM & ROLES

The people behind Disaster Shield AI



Full-Stack Developer

- React, Node.js
- MongoDB, REST APIs
- System Architecture



AI / ML Engineer

- Google Gemini AI
- NLP, Risk Modeling
- Data Pipelines



UI/UX Designer

- Figma, Framer Motion
- Glasmorphism UI
- User Research



GIS & Data Specialist

- Google Maps API
- GDACS, ReliefWeb
- Geospatial Analysis



DevOps & Security

- Cloud Deployment
- JWT Auth, HTTPS
- CI/CD Pipelines

MARKET OPPORTUNITY

Massive and growing addressable market

\$223B

Global Disaster
Management Market
by 2030

13.7%

CAGR growth rate
in disaster tech
sector (2023–2030)

5.2B

Smartphone users
who need reliable
disaster alerts

195+

Countries lacking
integrated real-time
disaster platforms

Disaster Shield AI is positioned to capture the underserved community resilience segment.

COMPETITIVE LANDSCAPE

Where Disaster Shield AI stands apart

Feature	Disaster Shield AI	FEMA App	ReliefWeb	GDACS
Real-Time GIS Map	🟢	🟡	🟡	🟡
AI-Powered Q&A	🟢	🟡	🟡	🟡
Survival Video Academy	🟢	🟡 🟡	🟡	🟡
Community Reporting	🟢	🟡	🟡	🟡
Offline Emergency Checklist	🟢	🟡	🟡	🟡
Multi-Hazard Coverage	🟢	🟡 🟡	🟡	🟡
Weather + Disaster Fusion	🟢	🟡	🟡	🟡
Open API / Developer Friendly	🟢	🟡	🟡 🟡	🟡

🟢 = Full Support | 🟡 🟡 Partial | 🟡 = Not Available

Built with trust at the core

JWT Authentication

- Stateless token-based auth
- Secure route protection
- Token expiry & refresh flow

HTTPS / TLS Encryption

- All traffic encrypted in transit
- MongoDB Atlas encrypted at rest
- Secure API key management

Input Validation

- Server-side schema validation
- Mongoose strict mode schemas
- XSS & injection protection

Environment Secrets

- All credentials via .env files
- No keys hardcoded in source
- Secrets rotated regularly

Compliance Ready

- GDPR-aligned data handling
- User consent for location data
- Minimal PII collection policy

Cloud Resilience

- MongoDB Atlas 99.95% uptime
- Auto-failover & backups
- Multi-region ready architecture

LIVE DEMO HIGHLIGHTS

Key screens from the running application



Interactive Disaster Map

Color-coded severity pins overlay real-time GDACS events onto Google Maps with impact radius circles and full telemetry popups.



Survival Academy

Filterable library of 10+ Red Cross, FEMA & CAL FIRE videos with category chips, difficulty tags, and step-by-step emergency guides.



AI Disaster Assistant

Gemini-powered chatbot answers natural-language questions, generates threat briefings, and offers survival advice 24/7.



Weather + Risk Dashboard

Hyperlocal OpenWeatherMap feeds fused with disaster risk scores show current conditions alongside threat probability heatmaps.



JOIN THE MISSION

Disaster Shield AI is open to collaboration, partnerships & investment.

Partner With Us

NGOs, governments & agencies looking to integrate real-time disaster intelligence

Contribute

Open-source contributors welcome:
github.com/Disaster-Shield-AI

Get In Touch

disaster.shield.ai@gmail.com
Reach out for pilots, demos & integrations

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